

ESC102 – Engineering Science Praxis II

Draft Request for Proposals

Team Name: TU13_T02_315

Team Members: Muhammed Ali, Michael Lin, Ziqi Yin, Hasnain Heryani

Date: 02/01/2026

1. Purpose of This Draft

2. Team Position

3. Community Overview

3.1. Community Definition

3.2. Stakeholders for this Project

3.2.1. Zain Ahmed: Snow Removal Business Owner (Primary Stakeholder)

3.2.2. Individual Snow Removal Business Owners in the GTA (Secondary Stakeholder)

3.2.3. Snow Removal Services Clients (Secondary Stakeholder)

3.2.4. Equipment retailers and Manufacturers (Tertiary Stakeholder)

3.2.5. Design Team (Primary Stakeholder)

4. Understanding the Community

5. Candidate Engineering Opportunities

5.1. Opportunity 1: Training New Sub-Contractors For Higher Efficiency Work using Technology

5.2. Opportunity 2: Ice Removal from Driveways

5.2.1. First approach: Targeted Thermal Weakening

5.2.2. Second approach: Repeated Impact Based Bond Weakening

5.3 Opportunity 3: Profitability and Quoting

6. Research Constructed to Date

6.1. Opportunity 1: Shovelling Path Optimization for New Contractors

6.2. Opportunity 2: Ice Removal from Driveways

6.2.1. First approach: Targeted Thermal Weakening

6.3. Opportunity 3: Profitability and Quoting

7. Next Steps

8. References List

9. Appendices

9.1. Appendix A: Meeting Transcript with Zain Ahmed for Snowshoveling

9.2. Appendix B: Calculations for Targeted Thermal Weakening

1. Purpose of This Draft

This draft document is a work-in-progress version of the team's final Request for Proposals (RFP). Its purpose is to document our progress to date in understanding the lived experience of a community, being small-scale snow removal businesses and in identifying possible opportunities to better support the community. It also captures ongoing secondary research and primary interactions with them. Additionally, it serves as a support to a meaningful in-studio consultation. Feedback from this will be used to refine the team's understanding, address assumptions, and clarify uncertainties in the framing of opportunities to aid the completion of our final RFP.

2. Team Position

As engineering students who have completed a semester of Praxis, we have gained extensive experience working on projects such as the Praxis I Design Project and CIV Bridge Project. Through these experiences, we have come to place strong value on clear communication and coordination, due to previously encountering challenges arising from misaligned expectations, uneven task distribution, and poor time management. Our current approach involves an emphasis on dividing work into smaller clearly defined tasks, assigning responsibilities with internal deadlines, and maintaining constant updates on our progress. Our task sheet, as seen in **Figure 1**, shows how as a team we have approached organization in our team.

Task	Assigned to	Due by	Done	Notes
Finish preliminary contact with communities	Everyone	Jan 22, 2026 (Thursday)	☒	
Contact MBA and Cousin for lawn mowing/snow plowing	Hasnain	Jan 22, 2026 (Thursday)	☒	
elderly's home (backup)	Frank	Jan 22, 2026 (Thursday)	☒	
Schedule Meeting with cousin for lawn mowing/snow plowing	Hasnain	Jan 26, 2026 (Tuesday)	☒	
Reply to Vision Loss Rehabilitation Canada	Muhammed Ali	Jan 27, 2026 (Tuesday)	☒	
Develop Team Position	Everyone	Jan 27, 2026 (Wednesday)	☒	
Finalize Community	Everyone	Jan 27, 2026 (Wednesday)	☒	Decided on snow shoveling community
Draft RFP Final Copy	Everyone	Jan 31, 2026 (Saturday)	☐	
Finalized Team Position	Frank	Jan 30, 2026 (Friday)	☒	
Research for all the opportunities	Everyone	Jan 31, 2026 (Saturday)	☐	
Figure out Opportunity 3	Everyone	Jan 30, 2026 (Friday)	☐	
Convert all secondary research into paragraph form	Hasnain	Jan 31, 2026 (Saturday)	☐	
Stakeholders		Jan 31, 2026 (Saturday)	☐	
Lived Experience	Hasnain	Jan 31, 2026 (Saturday)	☐	
Contextual Factors		Jan 31, 2026 (Saturday)	☐	
Opportunity 1 Research	Michael	Jan 31, 2026 (Saturday)	☐	

Figure 1: Task sheet with members and deadlines labelled and updated constantly

} Cut out the narrative
↳ What values we have
↳ how this aligns with Zain's business
↳ Put your work as reference evidence in Appendix

MAKE SURE TO HAVE BIASES and Assumptions

4 seasons yard work instead of business owner

stick with Zain as the community
talk to others only for NGOs

The importance of clarity and efficiency in communication is not just internal, but also shown externally, especially with the communities that we get in touch with for the RFP. This is mainly shown in two different aspects. On one hand, when we draft emails, we try to keep them professional and simple at the same time so that the reader can accurately capture the core ideas that we want to express. On the other hand, we have also considered communities that we have personal connections with so that early stage communication can be easier for both sides. For example, the community that we have chosen for the draft RFP is independent, small-scale snow removal contractors, where we had a personal connection with a contractor of such a business.

In addition to communication, we value the relevance of the chosen topic to our own lives. Choosing a topic that we are familiar with in our daily lives can give us solid foundational understandings of problems and makes it easier for us to identify opportunities and solutions. Most of our team members have lived in Canada for many years and deeply understand the trouble of removing snow in the winter. With a community that's closely related to our own lives, we have empathy and personal motivation to understand this community and we can draw on real experiences to test whether our interpretations make sense. Meanwhile, a potential risk caused by this is that we might over-generalize from our limited personal experiences and project our personal values or preferences onto the broader community. In order to prevent this, we must keep our claims objective and get in touch with more than one individual from the community to obtain various perspectives. Any perspective that we put into this as a claim, must have proper justification and evidence to minimize personal bias as much as possible and create something that would in turn benefit the broader community, not just those members that share our values.

As Praxis students, we are also going to apply the Praxis CTMFs we have learned in this course. Our Praxis I training provides structured approaches including Toulmin's argument structure, stakeholder analysis, requirements framework, diverging and converging, etc. These are all valuable systematic tools that can inspire meaningful engineering thinking and discussion throughout this project. However, we will definitely not be constrained by certain structures or concepts, but integrate these with what we are learning in Praxis II and our real-life experiences with engineering designs.

3. Community Overview

3.1. Community Definition

The target community consists of independent, small-scale snow removal business owners serving residential clients in the GTA as their shared interest. Zain Ahmed is one of these business owners and is our main point of contact and representative of this community. There is a conservative estimate of 300-500 individual or small snow removal contractors in the GTA, judging from active advertisement posts on Facebook

Regarding the community definition, we defined it as all the independent, small-scale snow removal businesses in the GTA and made Zain Ahmed the main point of contact and representative of this community. We were also considering another definition for the community, specifically making Zain Ahmed the owner of his snow removal business and all those he serves to be the target community. Which one of these definitions would be more appropriate for our RFP? What is your perspective on how the community should be defined?

Hasnain Heryani
2026-02-03 02:35

Also, does our current main point of contact for this community make sense? Since the community members do not actually interact with one another and only share an interest in clearing snow for residential clients and maximizing profit.

Hasnain Heryani
2026-02-03 02:51

Make
this
very
specific

NEED MORE EVIDENCE
HERE

Why does he remove snow?
to help people
we are structured and so is he

write from
the perspective
our client will
read it.

and Kijiji. Here are some key aspects or common interests that members of this community shares:

- Business entrepreneurship: They are responsible for all aspects of their business, including client acquisition through local paper or digital advertising, developing and adjusting pricing strategies, and personally investing in and maintaining their own equipment
- Operational challenges: They face similar challenges in winter in this area, which includes difficulty of travel in snowstorms, equipment breakdowns, need for manual work in freezing temperatures, and the unpredictability of income due to varying weather conditions.
- Client relationships: Instead of having a corporate network like larger companies, these individual business owners acquire new clients mainly by advertising locally, by paper or digitally. They contact existing customers directly and tend to develop stable and long-term loyal customers.
- Seasonal Nature: The operation time of the business only lasts 4-5 months. Therefore, they might have other businesses (such as lawn mowing) in the summer or additional income sources.
- Physical demand and safety: Members of this community usually have to do manual labour in extreme cold, leading to risk of frostbite or exhaustion. There are also risks of injury from slip or equipment operation or repetitive strain.

3.2. Stakeholders for this Project

(Rank them with influence)

3.2.1. Zain Ahmed: Snow Removal Business Owner (Primary Stakeholder)

As a business owner, his interest is to secure clients and maximize revenue, which involves providing high quality services. However, safety and efficiency are equally as important for both his clients and his sub-contractors.

3.2.2. Individual Snow Removal Business Owners in the GTA (Secondary Stakeholder)

Direct members of the target community. Their core interests are to secure clients and maximize revenue, which involve providing service of quality. Safety and efficiency of work are important considerations. Building and maintaining reputation among clients are also crucial.

→ This is 3rd
or fourth
↳ might benefit

3.2.3. Snow Removal Services Clients (Secondary Stakeholder)

Their main interest is to have a safe and clear driveway/walkway in front of their houses during snow season. They are looking for reliable service and reasonable pricing. Minimizing property damage is one important factor when they are choosing services.

3.2.4. Equipment retailers and Manufacturers (Tertiary Stakeholder)

Modifications to existing equipment or new equipment would have influences on their production process if applied. They aim to make profit by selling equipment and maintenance services to contractors. Quality, efficiency, and cost of equipment and tools are all significant factors taken into consideration for manufacturers.

3.2.5. Design Team (Primary Stakeholder)

The four team members in this Praxis II design team are also stakeholders for this project. As first year engineering science students, this project can help us gain more practice in engineering design, applying learned CTMFs and have a deeper understanding of engineering that we can bring into any future work.

4. Understanding the Community

In an online meeting with a member from the target community, we interviewed him around topics about his business routine, equipment operation, advertising methods, clients, difficulties and challenges in order to identify potential opportunities for improvement where engineering design can come into play. Talking to a member of the community has given us a deeper understanding of the community itself. Here are some inspiring aspects that the interviewee mentioned that influenced our diverging process. The full interview transcript can be found in Appendix A:

- He has transformed from advertising on paper at the beginning to mainly advertising digitally on platforms now.
- He has about 5-6 regular customers. Most of his customers are elderly who aren't physically suitable for shovelling snow by themselves and can afford snow removal services.
- As a university student who has to balance studies with this job and scheduling has been a problem for him during busy seasons.
- He has established a small network of subcontractors. Some of them can work in place of him whenever he is not free.
- Explaining setup and related knowledge required to new subcontractors can be complicated.
- He explained the difficulty of removing ice that sticks to the ground and claimed that it would be great if something can shatter the ice for him.
- He isn't planning to scale up his business because competition in the market would force him to lower his pricing and that he would have to pay more subcontractors. It is considered overall not worth it.
- Broken old equipment on site during work can be a real pain.

These lived experiences show that there are needs in various aspects of this job, which means that there is plenty of room for designs for improvement.

Page numbers?
Visuals? Integrals

Plan to meet in person

5. Candidate Engineering Opportunities

5.1. Opportunity 1: Training New Sub-Contractors For Higher Efficiency Work using Technology

As per the interview with Zain Ahmed, the owner of a small-scale snow shoveling business [See Appendix A], expresses difficulty with explaining and training new sub-contractors, especially with his busy university schedule. Specifically, he says: "then sometimes, like, if it's a new subcontractor, [...] having to explain, like, the equipment setup and how that works can be a little difficult, especially if they have questions" (22:49, Appendix A). Thus showing a difficulty in managing his time with his busy schedules. Additionally, he said that "I've learned just like the techniques to clear a driveway" (17:29, Appendix A), though it took him several years to learn this and so training new-subcontractors will speed up the learning process and thus result in higher quality work for the entire community.

Table 1: Impact of opportunity 1 on stakeholders

Stakeholder	Opportunity Impact
Zain Ahmed (Primary)	Saves more of his time, increases overall job efficiency, reduces costs
Sub-Contractors (Primary)	Faster learning curve, less time spent per job, less likely for workplace injuries
Clients (Primary)	Snow-removal business spends less time on their property, frees cars from snow faster
Other Snow Removal Businesses (Secondary)	Allowing other snow removal businesses across the GTA to gain access to new training methods can reduce their costs, provide higher quality work in less time
Equipment Manufacturers and Retailers (Tertiary)	Requires less maintenance as workers use equipment more efficiently, less accumulative equipment damage

5.2. Opportunity 2: Ice Removal from Driveways

This opportunity was identified from the discussion we had with the representative of our chosen community (transcript included in [Appendix A](#) Zain Ahmed: 29:44). It identifies the problem of ice that has been strongly bonded to the pavement. "There are some times where, like, I have trouble removing ice if it's just like, very thick, Especially at the snow bank at, like, the front of the driveways." Typical shovelling is not effective and this

↓
scope it down
type of working environment

} Make a profile
of the type of
ice? what kind
of driveway?
concrete, asphalt, interlocking?

Maybe
↑ Maybe posture / ergonomics
↳ Reduce risk
← Pitch it for risk, comfort and ergonomics
↓ type of workers using
↳ only male

method requires a lot of manual effort. The representative indicated that currently he simply salts it and lets it be, hence potential solutions for this issue would aim to completely remove the ice as an improvement on what the community already does.

Stakeholder	Opportunity Impact
Zain Ahmed (Primary)	Directly helps make the job easier and improve quality of service.
Sub-Contractors (Primary)	Decreases effort needed and makes the job easier
Clients (Primary)	Ensures the Ice is completely removed from their driveways.
Other Snow Removal Businesses (Secondary)	Directly helps make the job easier and improve quality of service.
Equipment Manufacturers and Retailers (Tertiary)	Potential solutions could serve as new products to produce.

Remember to add table number statement for this table

Hasnain Heryani
2026-02-03 03:06

5.2.1. First approach: Targeted Thermal Weakening

This approach would make use of a heating to melt the thin adhesive layer that exists on the ice-pavement boundary. The energy used to weaken the layer is similar to that of a household microwave (see [Section 6.1.2](#)) and can be easily achieved from a simple battery.

5.2.2. Second approach: Repeated Impact Based Bond Weakening

This approach makes use of repeated blows to the adhesive layer. A method that would slowly chip away at that layer until the ice can be separated from the pavement with relative ease. It will require some form of power supply to function.

5.3 Opportunity 3: Profitability and Quoting

One of the major business considerations would be determining how much to quote each customer based on square feet of the areas that require shovelling.

Residential properties vary enormously by area. For example, a suburban Mississauga home might have 75m of linear shoveling, while a typical downtown Toronto townhouse might have about 15m. Additionally, there is a more hidden cost of gas money required to get to a certain location. Independent contractors often face profitability calculation problems when they receive certain inquiries. Taking these two things into consideration, it allows for contractors to get a better idea on how lucrative a certain job opportunity could be.

The unique market position of independent contractors contribute to the significance of this need. Large commercial companies have dedicated route optimization software and

dispatch coordinator, while independent business owners don't usually have access to required algorithms. When small business owners try to build viable business or expand their business, given high gas prices in the GTA and unknown client demographics, they often lack tools to assess profitability at scale.

Stakeholder	Opportunity Impact
Zain Ahmed (Primary)	Decreases friction in orders, increases sales and decreases manual calculation time.
Sub-Contractors (Primary)	More transparent and immediate knowledge transfer of job type and payout.
Clients (Primary)	Transparent Pricing and Immediate rates likely to build more trust and lead to more clients actually taking up the service.
Other Snow Removal Businesses (Secondary)	Decreases friction in orders, increases sales and decreases manual calculation time.
Equipment Manufacturers and Retailers (Tertiary)	Possible software for them to integrate into their current projects.

Remember to add table number statement for this table

Hasnain Heryani
2026-02-03 03:06

6. Research Constructed to Date

6.1. Opportunity 1: Shovelling Path Optimization for New Contractors

The labour of snow shovelling can be quite expensive. For example, the City of Toronto allocated \$140 million in 2025 for snow shovelling operations [7]. Even a slight decrease in the efficiency of people manually shovelling snow can be costly for subcontractors. Quicker and more efficient jobs can allow subcontractors to more easily continue with their daily tasks after work. As stated in the interview with Zain Ahmed, a busy lifestyle outside of snow shovelling jobs makes it profitable for subcontractors to be able to more efficiently finish jobs. Additionally, the physical strain of shovelling snow can take a toll for longer term subcontractors who may do multiple shifts a week. If someone were to be shovelling too rigorously for an extended amount of time, they might have a heart rate above a safety level of approximately 180 beats per minute [8]. By allowing subcontractors to be able to track their inefficiencies, they can better learn on how to conserve their energy and use it more efficiently. Thus, we are able to decrease the risk of injury while also saving subcontractors from doing more redundant strenuous work.

Reference Design:

Our potential solution here might be too experimental, doesn't really have practical effects. How may experimental solutions that aren't extremely practical affect us?

Michael Lin
2026-02-03 02:48

Needs to be integrated into everything

A related design that serves as a precedent for this opportunity is fleet telematics, a field led by the Canadian company Geotab. They specialize in tracking fleets of trucks by converting raw vehicle movement data into trackable efficiency scores [6]. Fleet telematics can store data based on driving behaviour including distance, time, acceleration, and any sharp turns. This is done in order to optimize efficiency while saving business owners money and time. Geotab is able to eliminate production inefficiencies by determining if a driver is taking a redundant route or driving in a manner that causes premature damage to the vehicle. Overall, fleet telematics and the work that Geotab has done is similar to the opportunity that is being investigated here.

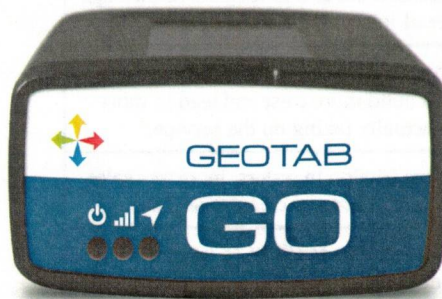


Figure 2: Physical Geotab device

6.2. Opportunity 2: Ice Removal from Driveways

6.2.1. First approach: Targeted Thermal Weakening

- **Core Principal**

- Ice sticks to pavement through mechanical interlocking (when ice goes into the cracks of the pavement and fills surface pores) and adhesive bonds. Fully breaking or melting the ice is likely to take too much time and energy. Hence, for this opportunity it is best to find an approach that gets rid of the adhesive layer [1].

- **First approach: Targeted Thermal Weakening**

- Localized heating at the ice-pavement boundary, breaking adhesion, does not fully melt the ice.
- Reference Design for this kind of issue is the POWERFIST 130W Electric Hot Knife [8] (See Figure 3)

With regards to this opportunity, the reference designs used have functions that are analogous to features that could be used on potential solutions. However, the designs themselves are used for different purposes. Would it be better for the final RFP to include reference designs that already exist to remove ice.

Muhammed Ali Bhanji
2026-02-03 02:52



Figure 3: POWERFIST 130W Electric Hot Knife

- Can be delivered via heated metal element.
- This sort of tool highlights realistic possibilities as less power is needed
- Using the heating principle to melt the adhesive layer, not fully melt the ice. The energy used to weaken ice adhesion is similar to a household microwave.
- See calculation in [Appendix B](#).
- Key challenges with reference design:
 - The space needed to store the battery on the tool itself, could increase the weight by a lot, making it heavy and inconvenient.
 - Safety risk exists with heating elements.
 - Ensuring the heating element causes no damage to pavement.

- **Second approach: Repeated Impact Based Bond Weakening**

- A good reference design for this is the DEWALT DCH172B ATOMIC 20V MAX 5/8" Brushless Cordless SDS Plus Rotary Hammer [9] (See Figure 4)



Figure 4: DEWALT DCH172B ATOMIC 20V MAX 5/8" Brushless Cordless SDS Plus Rotary Hammer

- This sort of rotary hammer can easily be used for chiseling through harder tough materials like concrete and stone. It uses pistons to deliver high-impact repeated force, significantly more powerful than a standard hammer.



- Chiseling away at the adhesive layer of a material like ice would require far less power due to the brittle nature of ice.
- Requires less sustained user force.
- Key challenges:
 - Limiting damage to concrete & asphalt.
 - Tool Stability.

6.3. Opportunity 3: Profitability and Quoting

- The essential need is to auto-calculate and present the user with a quote based on the distance travelled and area of the driveway.
- Based on need presented by community representative in [Appendix A](#) “There have been in the past, especially when the business was new, where I took on jobs that were, like, too big, that, like, I just couldn't do, and it ended up taking a lot longer than I thought it would, and I ended up undercharging. But other than that, I can't think of any.” (23:45)
- The reduced friction with an automatic method and transparent pricing would boost sales for the service
- This kind of price transparency at the time of the checkout increases how many customers actually convert [7].

7. Next Steps

This draft represents our team's progress through secondary research and preliminary analysis. Moving on, here are some steps that we will take towards the final outcome of the project:

- Refine and validate each opportunity and further explore the solution space.
- Continue reaching out to the community and discuss the opportunities and potential solutions with them to validate our direction.
- Do more iterations of converging and diverging to more clearly define our current options.
- Conduct more in-depth secondary research on the design opportunities that we have identified to generate early-stage prototyping ideas for designs.
- Gain insight of feasibility of potential solutions
- Try to hold interviews with other small-scale snow removal contractors and some of their clients.

With regards to the research on Opportunity 3, we realized there is not much secondary research that we could think of that directly relates to the scope of this opportunity. Do you have any suggestions for different angles we could take in approaching the research aspect of this opportunity.

Muhammed Ali Bhanji
2026-02-03 02:44

In terms of this opportunity, since the community is focusing on the owners of snow removal businesses, one of their main values includes profit maximization. So would it be appropriate to address profit maximization as a potential opportunity, specifically trying to reduce friction, uncertainty

Hasnain Heryani
2026-02-03 02:39

Additionally, we had a concern related to this opportunity because of something he said during our interview with him. He said that the majority of this clients are elderly and tech illiterate, so if we were to approach this opportunity from a technology perspective, where his clients were to use a software to book

Hasnain Heryani
2026-02-03 02:45

What are your opinions on our next steps? Should we also address anything?

Hasnain Heryani
2026-02-03 03:04